

ST PETERS, PE, Canada

WMO: 713100

Lat: 46.4503N

Lon: 62.5758W

Elev: 30

StdP: 100.97

Time Zone: -4.00 (NAA)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.7	-15.4	-21.9	0.5	-16.7	-19.7	0.7	-14.7	13.2	-7.4	12.2	-6.7	5.1	240	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	27.6	21.0	26.1	20.0	24.5	19.3	22.2	26.1	21.2	24.5	20.3	23.3	5.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.7	15.4	24.4	20.0	14.7	23.6	19.1	13.9	22.4	65.2	26.4	61.7	24.5	58.6	23.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.9	8.8	-20.6	30.2	1.8	1.0	-21.8	30.8	-22.9	31.4	-23.9	31.9	-25.2	32.6		
(5)			-21.0	23.7	1.7	0.8	-22.2	24.3	-23.2	24.8	-24.1	25.2	-25.4	25.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-5.2	-6.0	-2.6	3.3	8.7	14.0	19.2	18.9	15.1	9.5	4.2	-1.5	(6)	
	DBStd	9.63	5.54	4.98	4.34	4.05	4.03	3.88	3.05	2.60	3.40	3.64	4.25	4.75	(7)	
	HDD10.0	2193	469	448	392	208	76	7	0	0	2	53	182	356	(8)	
	HDD18.3	4422	728	681	650	452	300	139	27	24	108	273	425	614	(9)	
	CDD10.0	928	0	0	1	6	34	127	285	276	154	39	7	0	(10)	
	CDD18.3	115	0	0	0	0	1	9	53	42	10	1	0	0	(11)	
	CDH23.3	592	0	0	0	1	9	58	302	184	37	2	0	0	(12)	
	CDH26.7	81	0	0	0	0	1	4	49	23	3	0	0	0	(13)	
(14) Wind	WSAvg	4.4	5.2	4.8	4.7	4.4	4.1	3.8	3.6	3.5	4.1	4.4	4.9	5.2	(14)	
(15) Precipitation	PrecAvg	1209	104	86	90	84	81	82	87	99	107	127	118	135	(15)	
	PrecMax	1563	249	150	148	145	169	155	171	217	215	267	207	280	(16)	
	PrecMin	982	36	18	34	38	48	35	21	30	28	46	54	67	(17)	
	PrecStd	149	42	33	30	27	27	28	34	54	52	54	35	44	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	9.7	11.3	20.0	24.5	27.2	29.6	28.9	26.6	22.1	18.0	13.1	(19)	
		MCWB	11.8	8.8	7.2	14.5	17.2	19.2	22.6	21.6	21.1	17.5	15.7	12.1	(20)	
	2%	DB	7.7	5.8	7.7	16.1	21.1	25.1	28.1	27.1	24.1	19.2	15.4	10.0	(21)	
		MCWB	7.1	4.4	5.6	12.4	14.5	18.5	21.6	20.8	19.6	16.0	14.0	9.1	(22)	
	5%	DB	4.6	2.8	5.3	12.6	18.4	23.0	26.7	25.4	22.5	16.9	13.3	7.4	(23)	
		MCWB	3.5	1.5	3.5	9.5	13.1	17.5	20.5	19.8	18.4	14.6	12.0	6.6	(24)	
	10%	DB	2.0	0.6	3.2	9.9	15.8	20.8	25.0	23.8	20.7	15.1	10.8	4.9	(25)	
		MCWB	1.1	-0.3	1.5	7.3	11.8	16.4	19.7	19.1	17.6	12.9	9.4	4.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	8.7	8.7	16.0	18.4	21.7	23.4	23.0	22.0	19.0	16.3	12.2	(27)	
		MCDB	13.0	9.7	10.1	18.7	23.2	25.3	28.3	26.7	25.3	20.9	17.8	12.8	(28)	
	2%	WB	7.1	4.6	6.0	12.9	16.2	20.1	22.5	21.7	20.5	16.5	14.2	9.2	(29)	
		MCDB	7.5	5.5	7.1	15.4	19.3	23.1	26.9	25.3	23.3	18.2	15.3	9.9	(30)	
	5%	WB	3.7	1.7	3.6	10.3	14.0	18.4	21.6	20.8	19.3	14.9	12.2	6.7	(31)	
		MCDB	4.5	2.7	5.1	12.1	16.8	21.6	25.2	23.9	21.5	16.8	13.1	7.3	(32)	
	10%	WB	1.3	-0.2	1.6	7.3	12.1	16.7	20.6	19.9	18.1	13.2	9.5	4.1	(33)	
		MCDB	1.9	0.5	3.1	9.6	15.3	20.3	23.7	23.0	20.1	14.9	10.6	4.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	7.2	6.9	7.5	8.6	8.2	8.3	7.9	7.9	6.6	6.3	6.0	(35)	
		MCDBR	10.8	9.9	10.1	12.8	13.8	11.9	10.9	10.4	10.2	9.4	9.6	9.5	(36)	
	5% WB	MCWBR	10.9	8.9	8.5	9.5	8.5	6.7	5.7	5.5	6.5	7.3	9.3	9.8	(37)	
		MCWBR	11.0	10.0	9.6	12.0	11.9	10.1	9.1	8.8	8.6	8.6	9.5	9.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.299	0.289	0.319	0.349	0.368	0.383	0.409	0.397	0.358	0.338	0.319	0.295	(40)	
		taud	2.360	2.289	2.296	2.403	2.429	2.403	2.370	2.404	2.502	2.522	2.490	2.458	(41)	
	Edn at Noon		793	884	906	908	898	882	853	850	859	825	771	764	(42)	
		Edh at Noon	73	95	110	110	111	115	117	109	91	77	66	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.02	1.72	3.10	4.20	5.13	5.59	5.68	4.98	3.80	2.33	1.19	0.80	(44)	
	RadStd		0.06	0.25	0.31	0.38	0.45	0.41	0.38	0.30	0.33	0.24	0.15	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.40	N/A	N/A	N/A	N/A

Nomenclature: See separate page